



Characterization Summary of the 20 CalVision BGO Crystals

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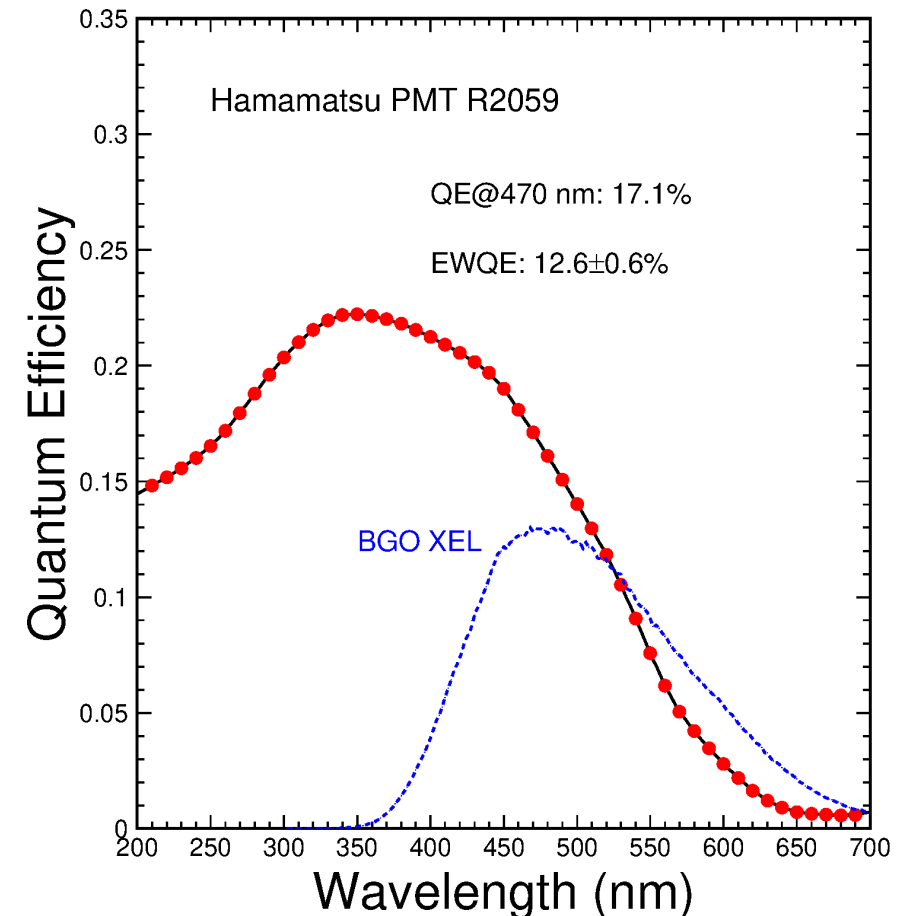
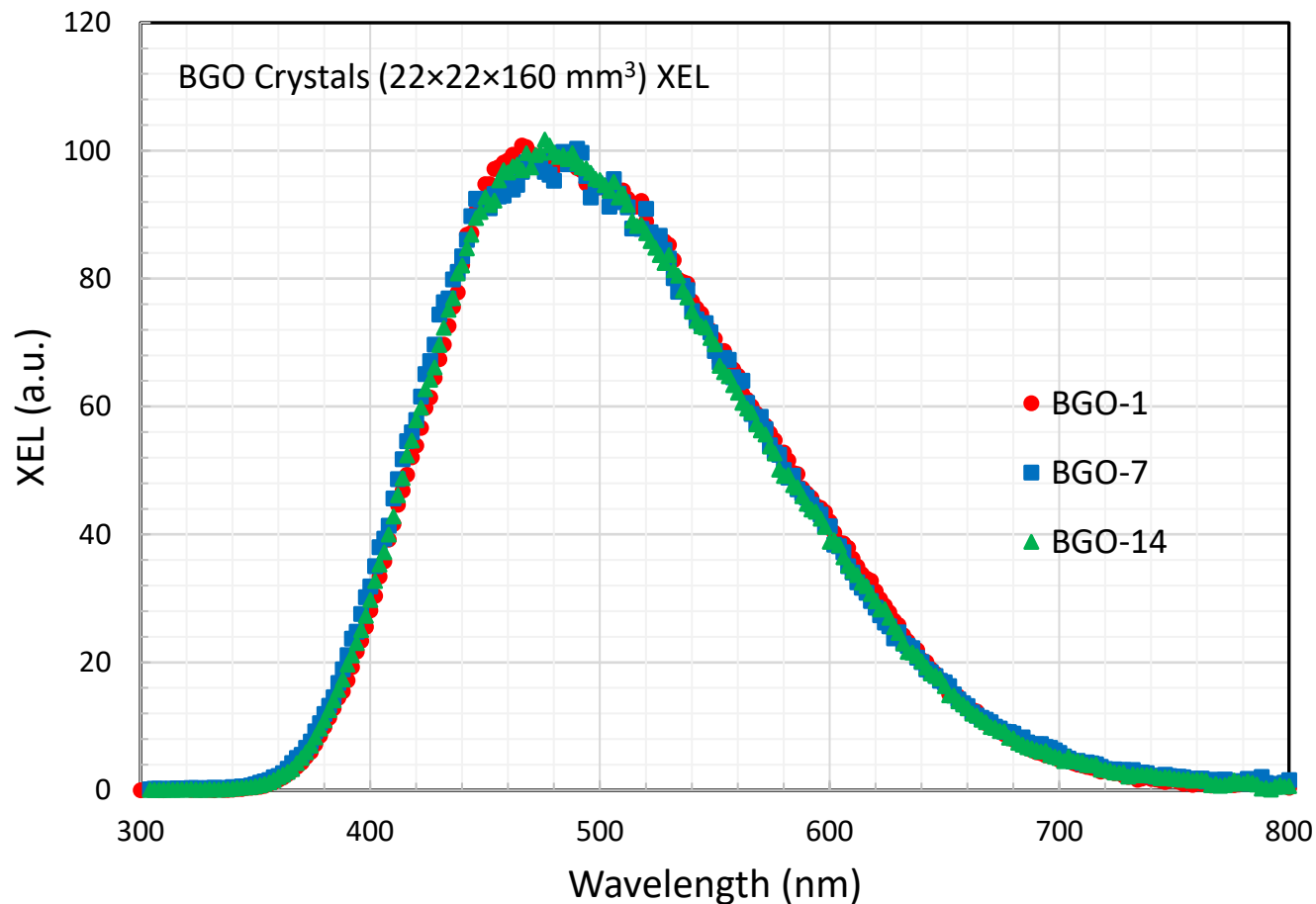
Twenty BGO Crystals ($22\times22\times160\text{ mm}^3$)



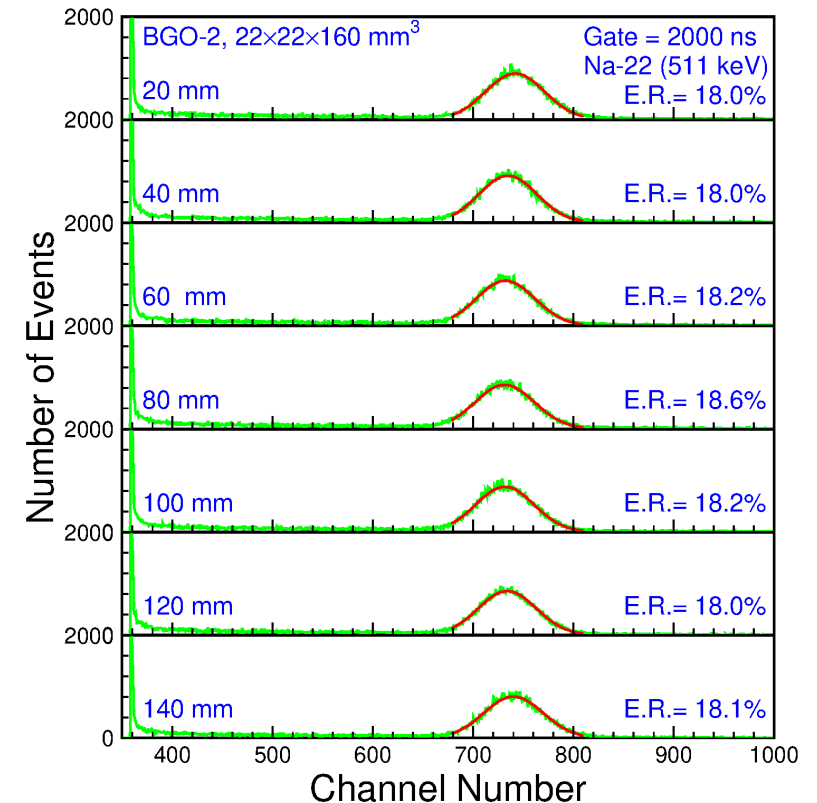
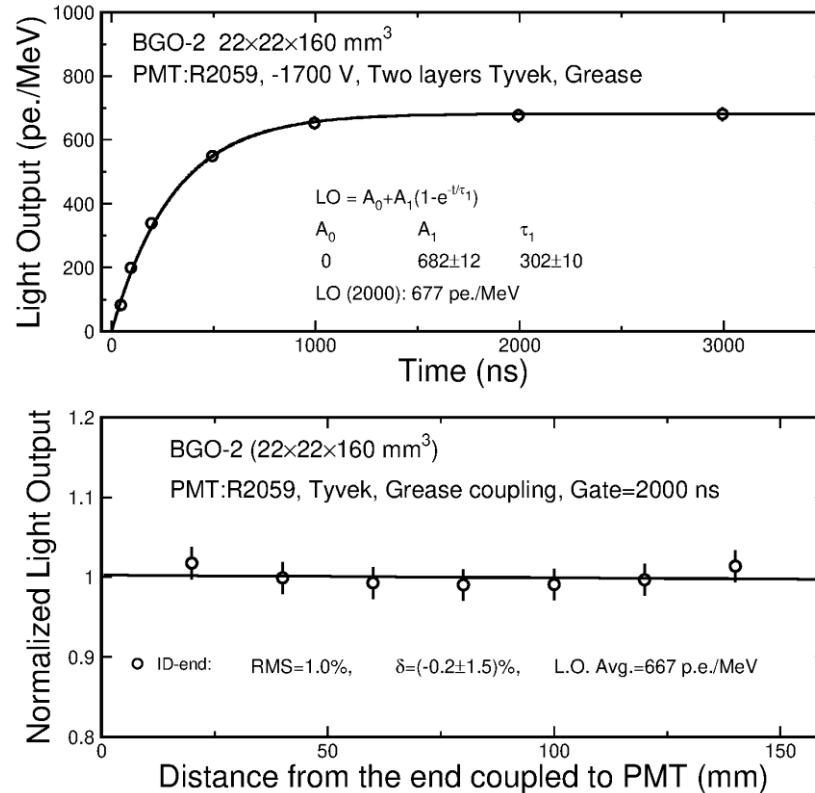
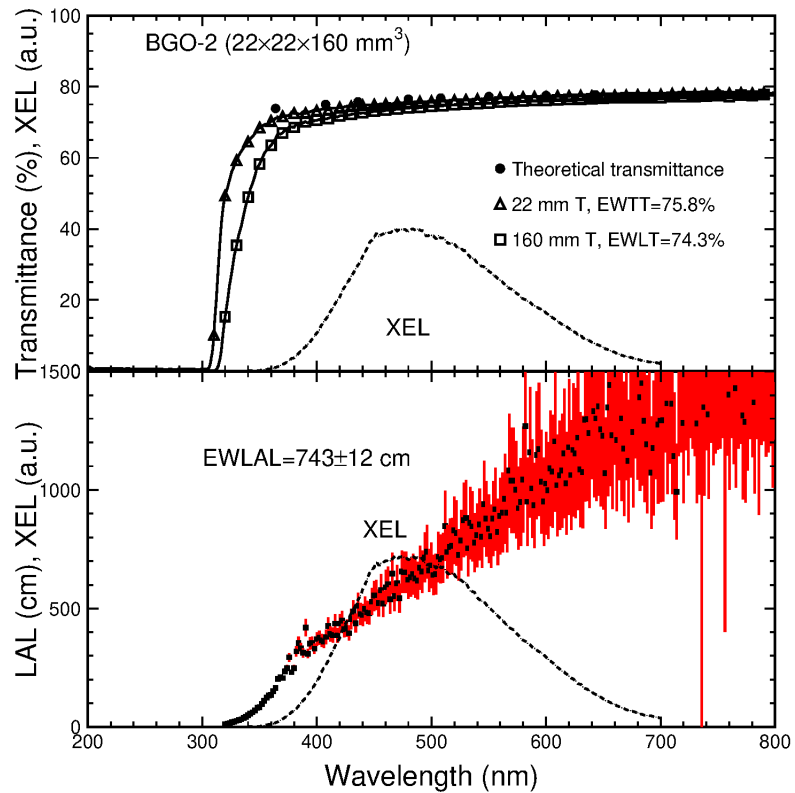
- The 20 Caltech BGO of $22\times22\times160\text{ mm}^3$ crystals were delivered to Caltech on Jan. 13, 2025.
- Measurements at room temperature:
 - X-ray excited luminescence (XEL),
 - Longitudinal/Transverse transmittance (LT/TT),
 - Light Output (LO) & Decay Time(τ), Light Response Uniformity (LRU), and Light Yield (LY) with Emission Weighted Quantum Efficiency (EWQE) taken out.
- Six crystals were fully measured and reported on Feb. 20 and Mar. 20, 2025.
- All other crystals are measured for LT/TT, decay, LO and LRU and summarized in this report.

XEL and EWQE

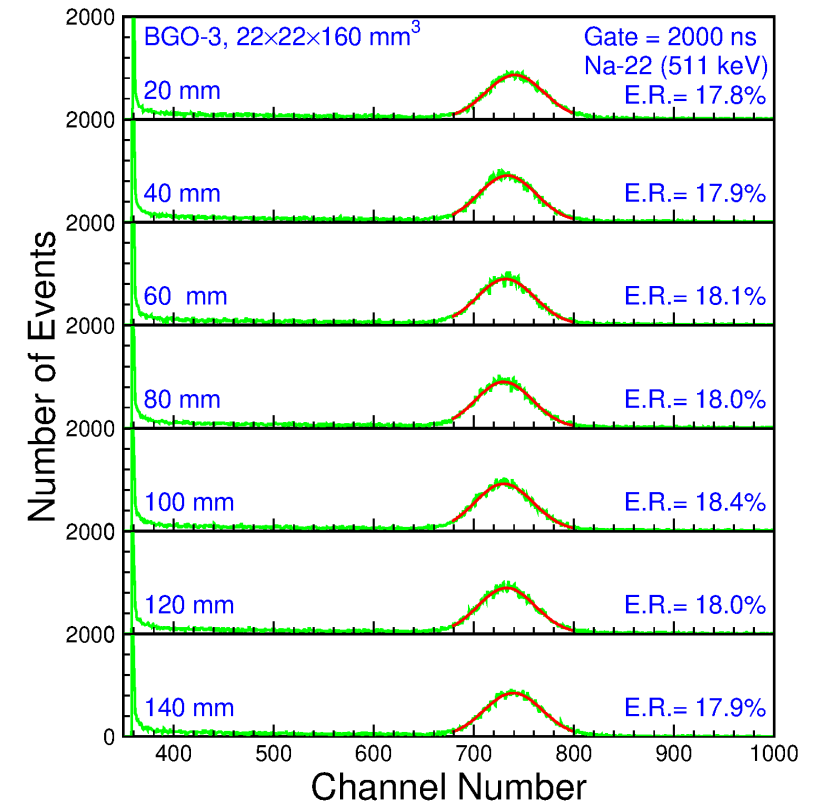
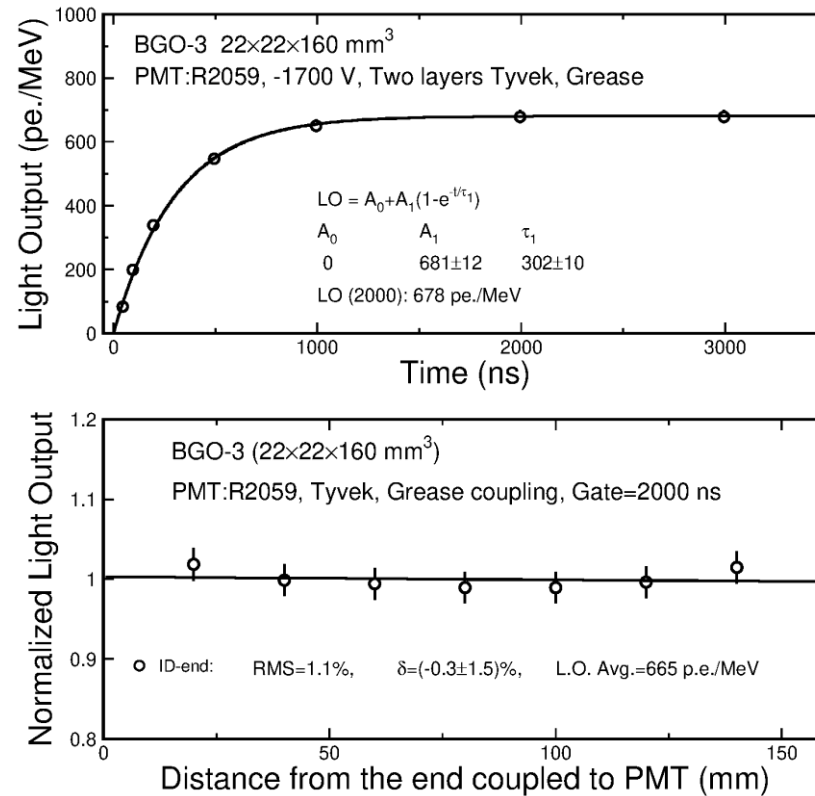
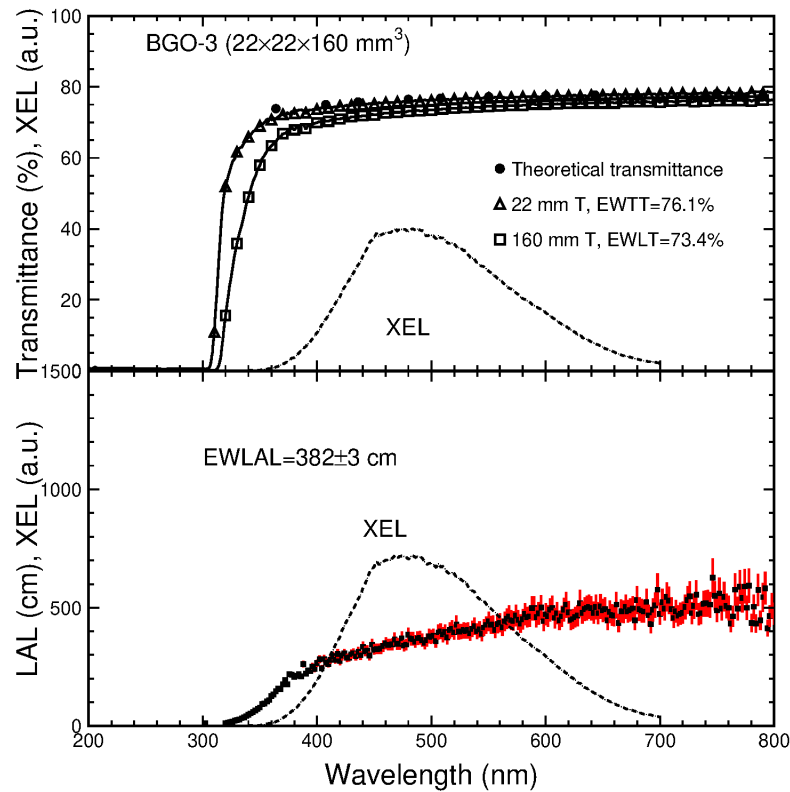
- The X-ray excited luminescence (XEL) was measured for 6 BGO crystals, the emission spectra are consistent.
- The average of XEL spectra is used to calculate emission weighted quantum efficiency (EWQE) of the photo-detector (PMT R2059) in the light output measurement.



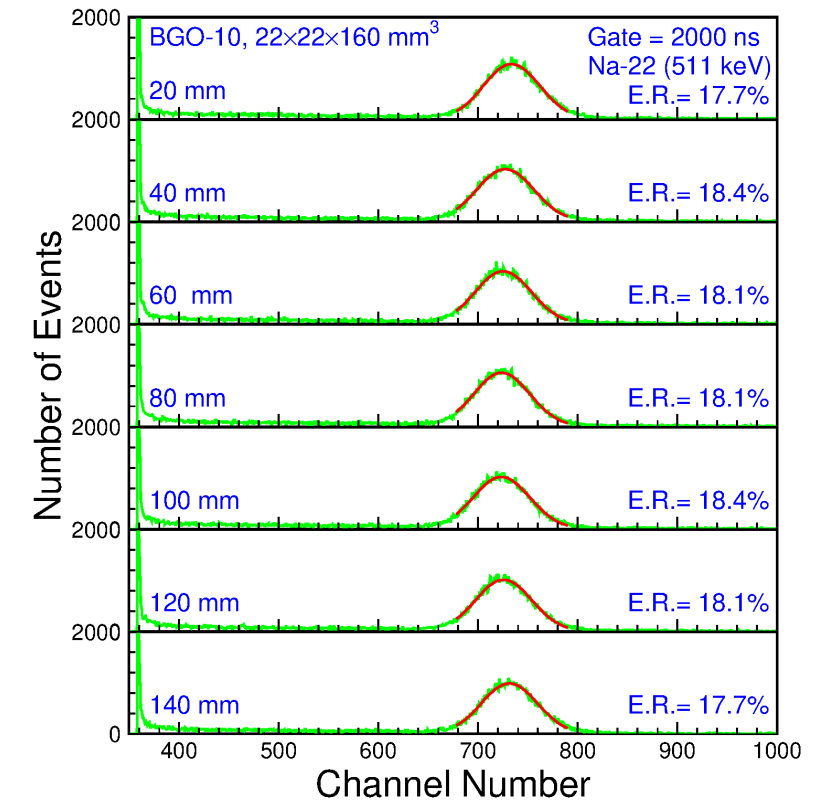
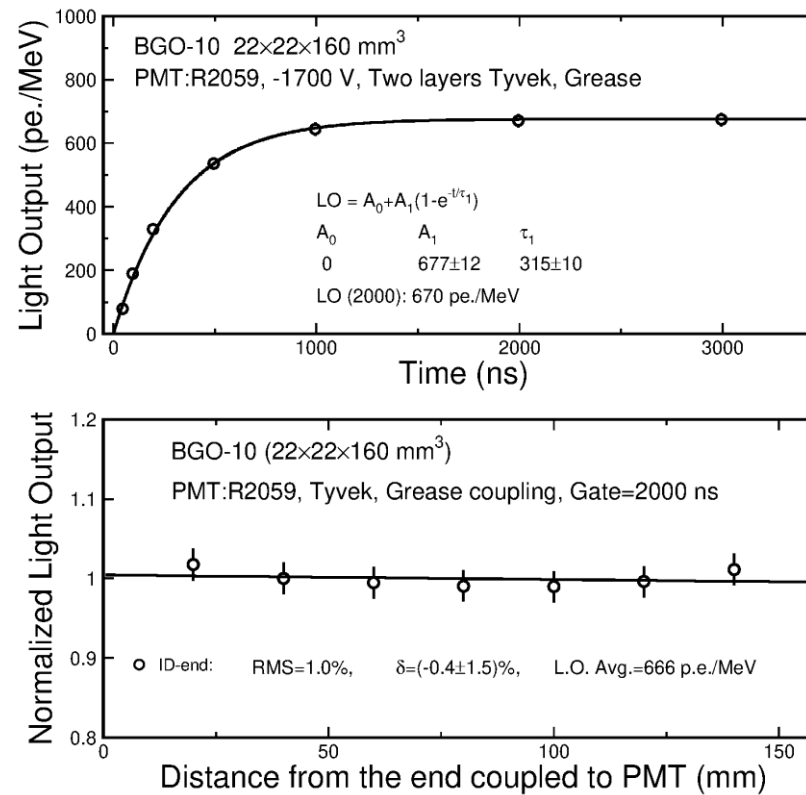
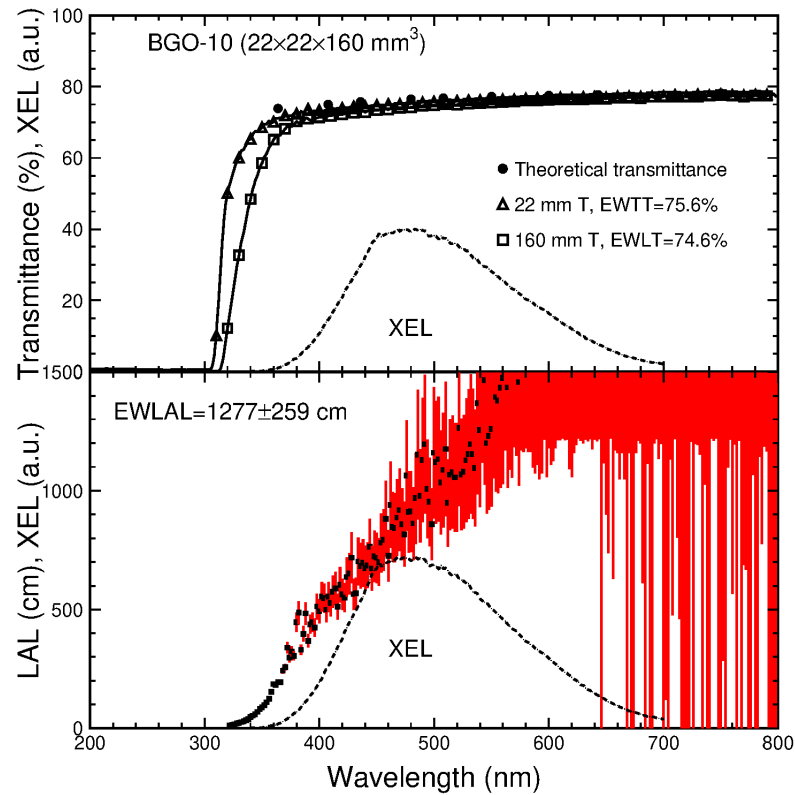
T, LAL, Decay, LO, LRU and ER (BGO-2)



T, LAL, Decay, LO, LRU and ER (BGO-3)



T, LAL, Decay, LO, LRU and ER (BGO-10)

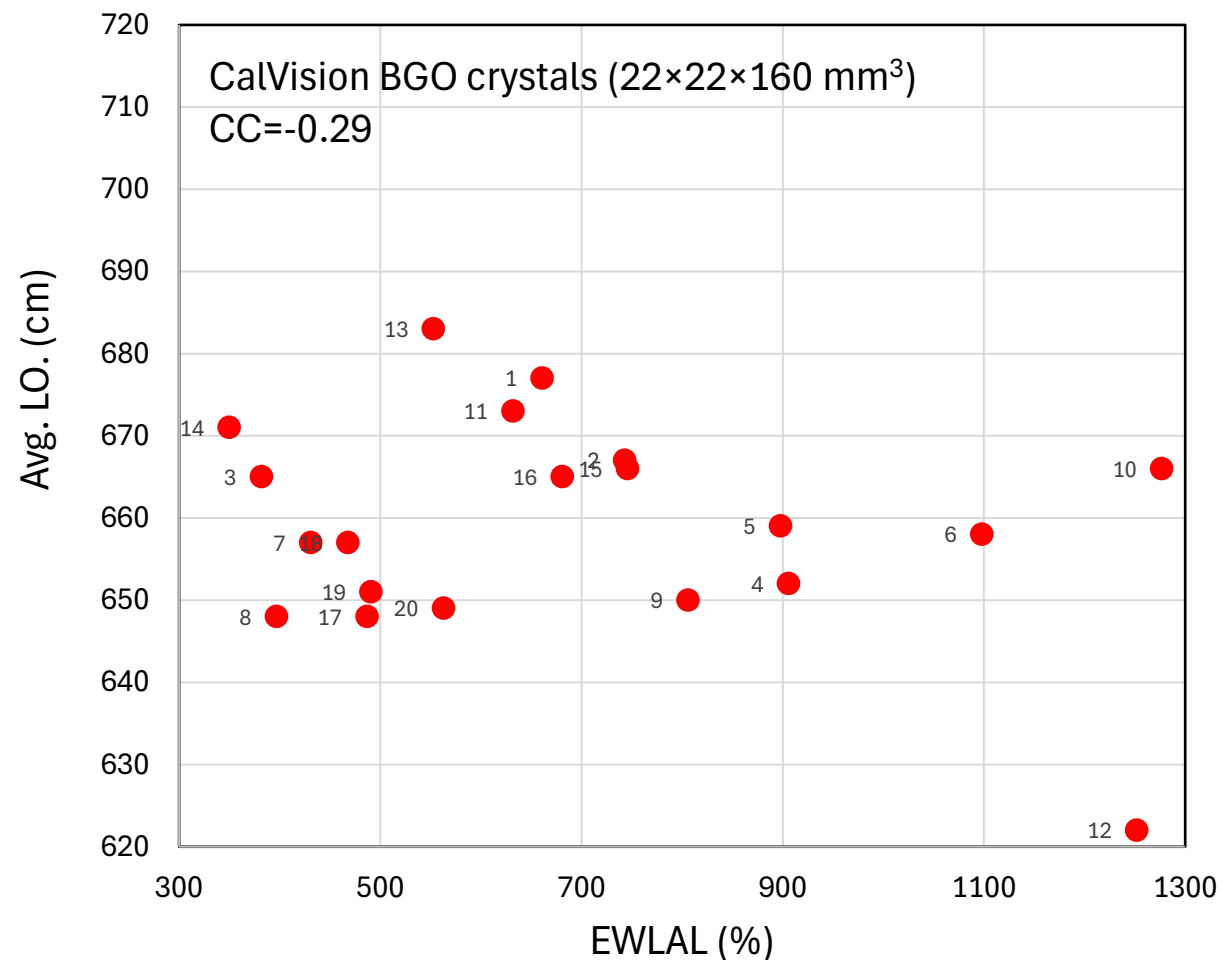
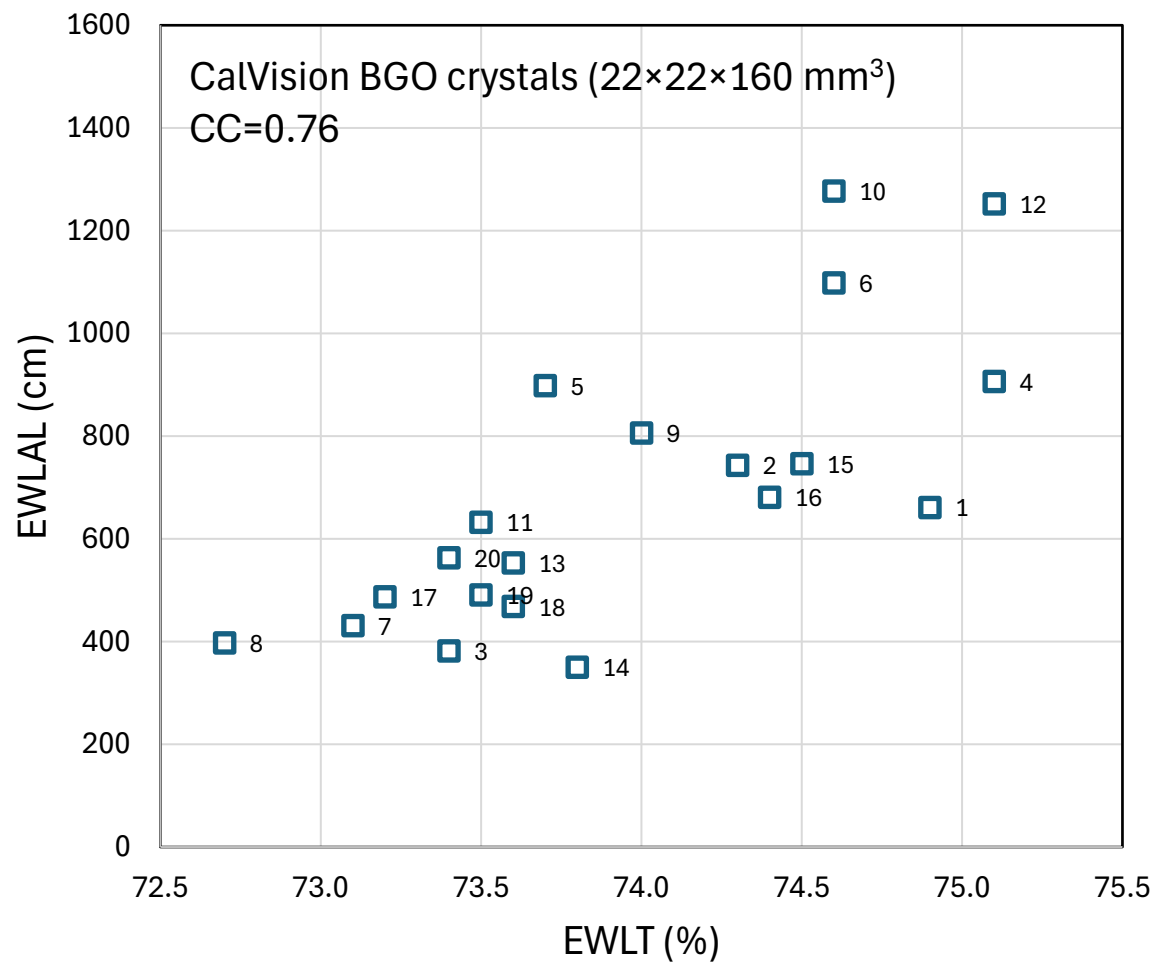


Summary

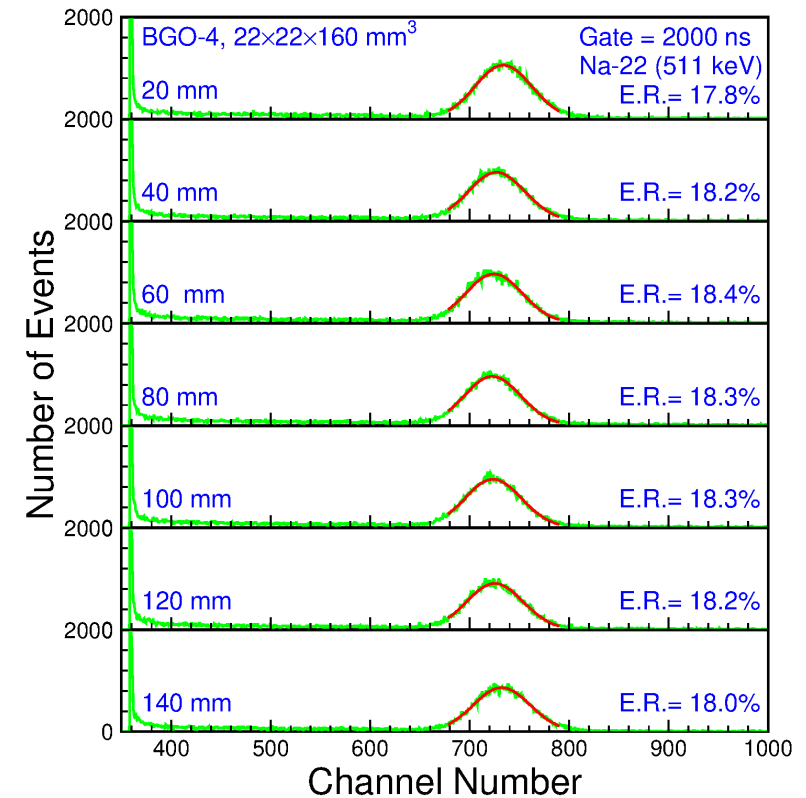
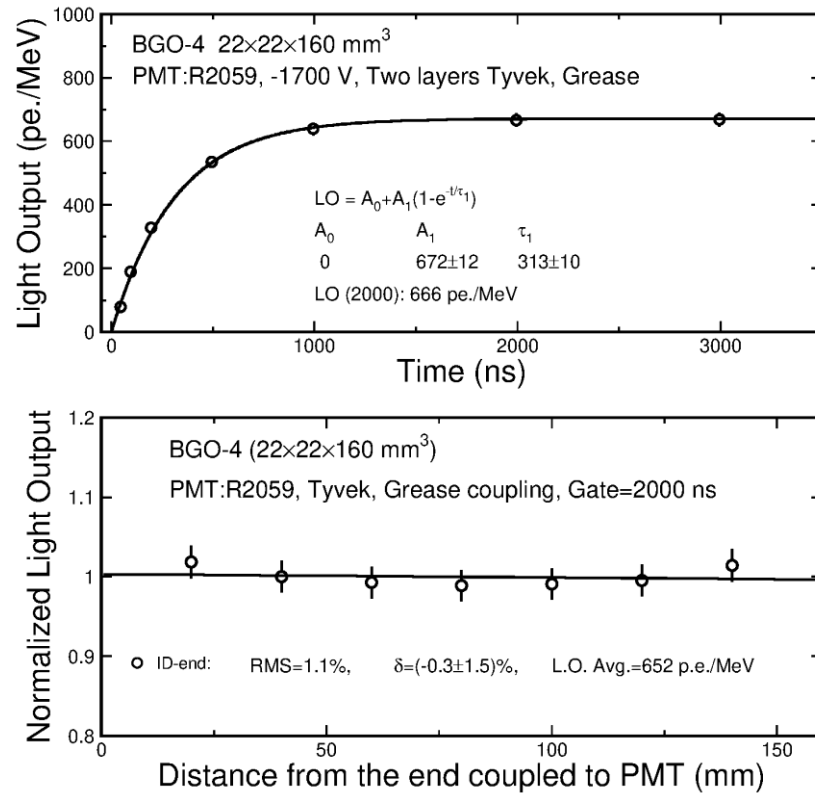
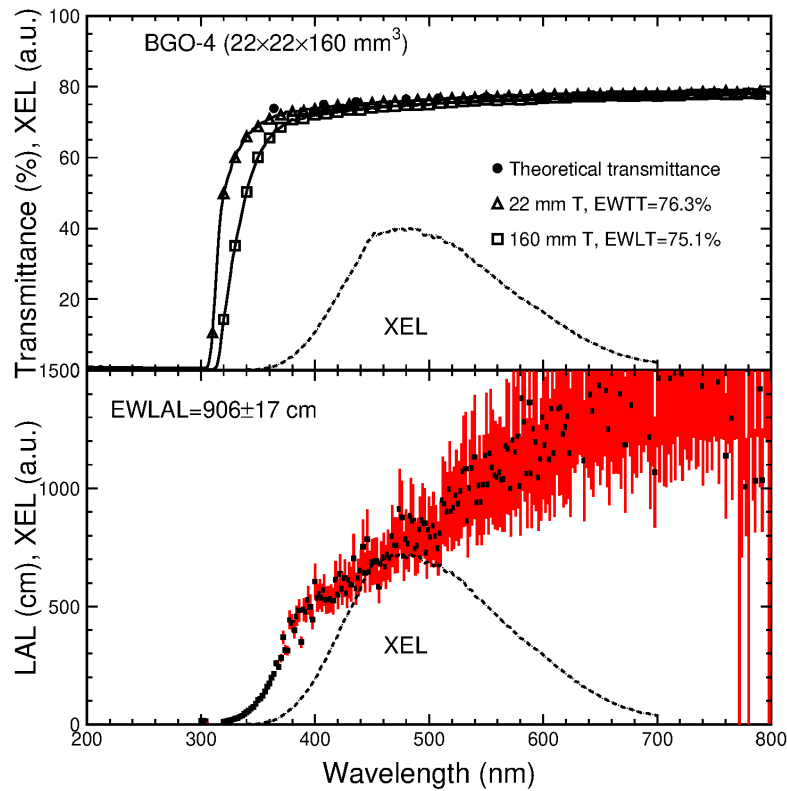
ID	EWLT (%)	EWLAL (cm)	Decay (ns)	LO avg. (p.e./MeV)	LO avg. (ph./MeV)	LRU (%)	Avg. ER. (511 KeV) (%)
1	74.9	661	308	677	5.4E+03	1.2	17.2
2	74.3	743	302	667	5.3E+03	1.0	18.2
3	73.4	382	302	665	5.3E+03	1.1	18.0
4	75.1	906	313	652	5.2E+03	1.1	18.2
5	73.7	898	315	659	5.2E+03	0.9	17.9
6	74.6	1098	297	658	5.2E+03	1.2	17.6
7	73.1	431	287	657	5.2E+03	1.1	17.6
8	72.7	397	309	648	5.1E+03	1.1	18.3
9	74.0	806	294	650	5.2E+03	1.0	18.2
10	74.6	1277	315	666	5.3E+03	1.0	18.1
11	73.5	632	314	673	5.3E+03	1.0	17.9
12	75.1	1252	298	622	4.9E+03	1.1	18.0
13	73.6	553	306	683	5.4E+03	0.9	17.9
14	73.8	350	305	671	5.3E+03	1.1	17.5
15	74.5	746	294	666	5.3E+03	0.9	18.2
16	74.4	681	297	665	5.3E+03	1.1	18.1
17	73.2	487	308	648	5.1E+03	1.0	18.4
18	73.6	468	313	657	5.2E+03	1.1	18.1
19	73.5	491	308	651	5.2E+03	1.1	18.2
20	73.4	563	297	649	5.2E+03	0.9	17.5
Avg.	74.0	691	304.1	659.2	5.2E+03	1.0	18.0
Std.	0.7	278	8.1	13.4	106	0.1	0.3
Std/Avg. (%)	0.9	40	2.7	2.0	2.0	9.0	1.8

Discussion

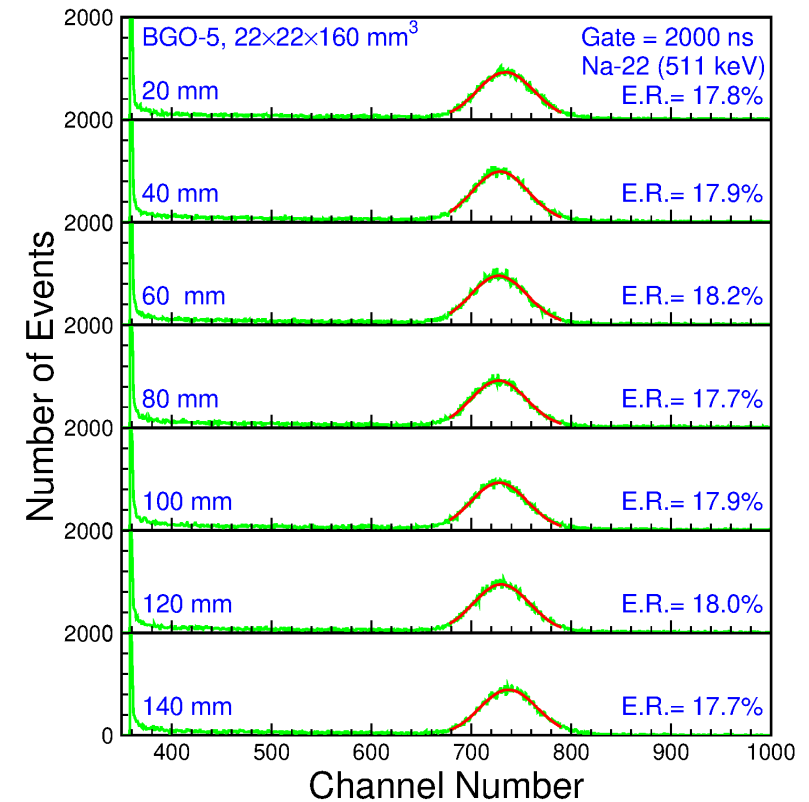
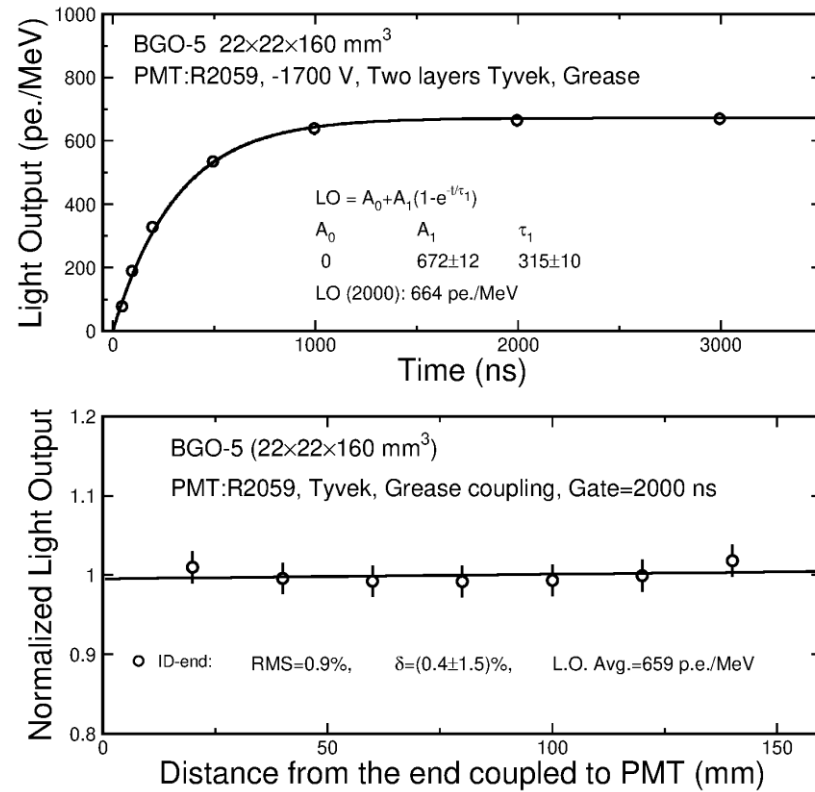
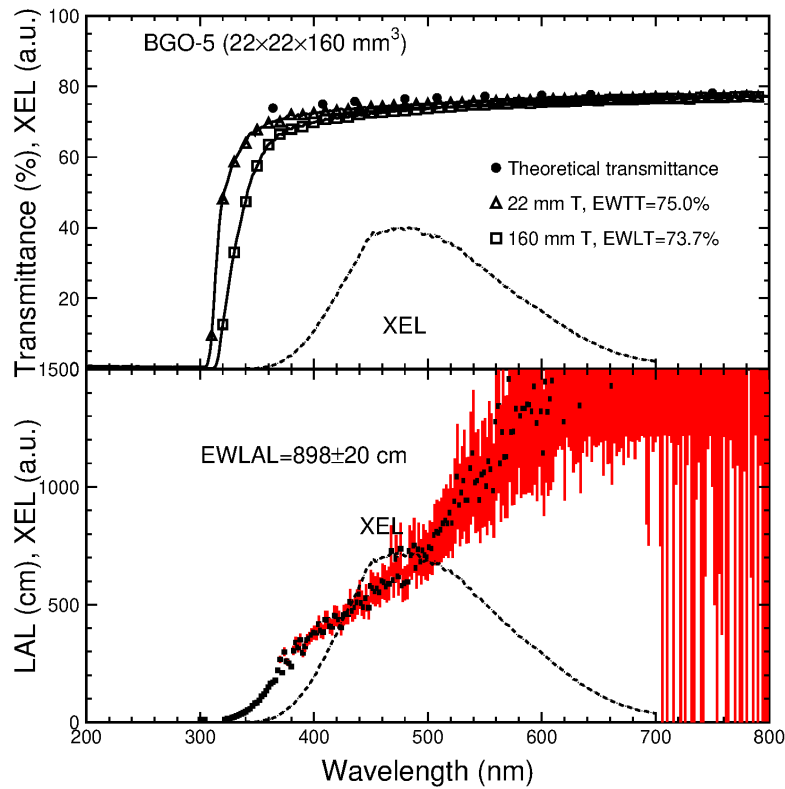
- Slight correlation between the emission weighted light attenuation length and longitudinal transmittance.
- No correlation is observed between the light output (and light response uniformity) and the emission weighted light attenuation length, indicating transmittance would not be a limiting factor.



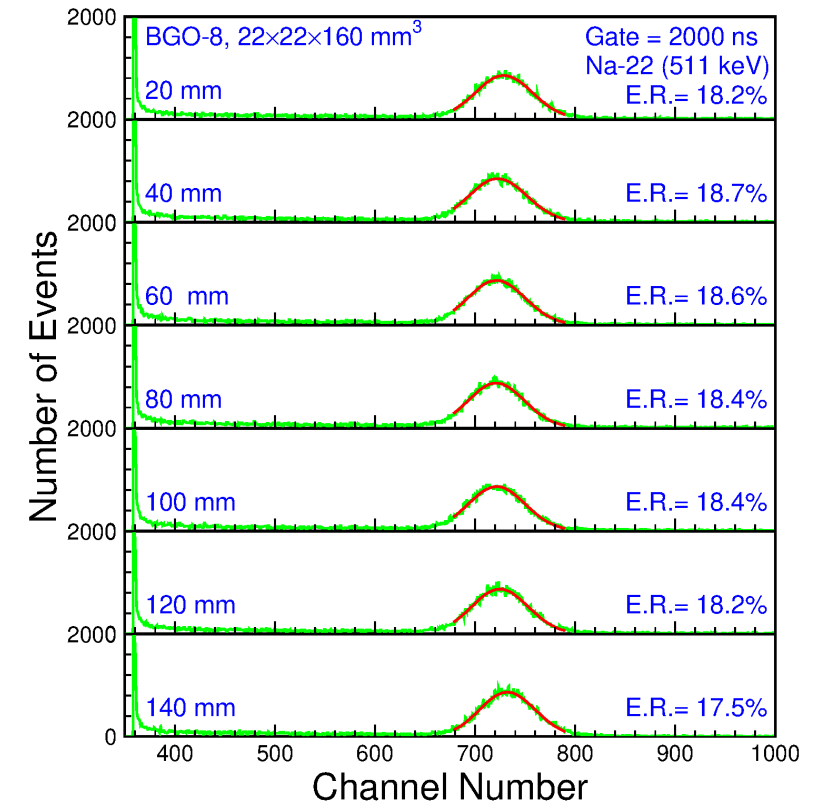
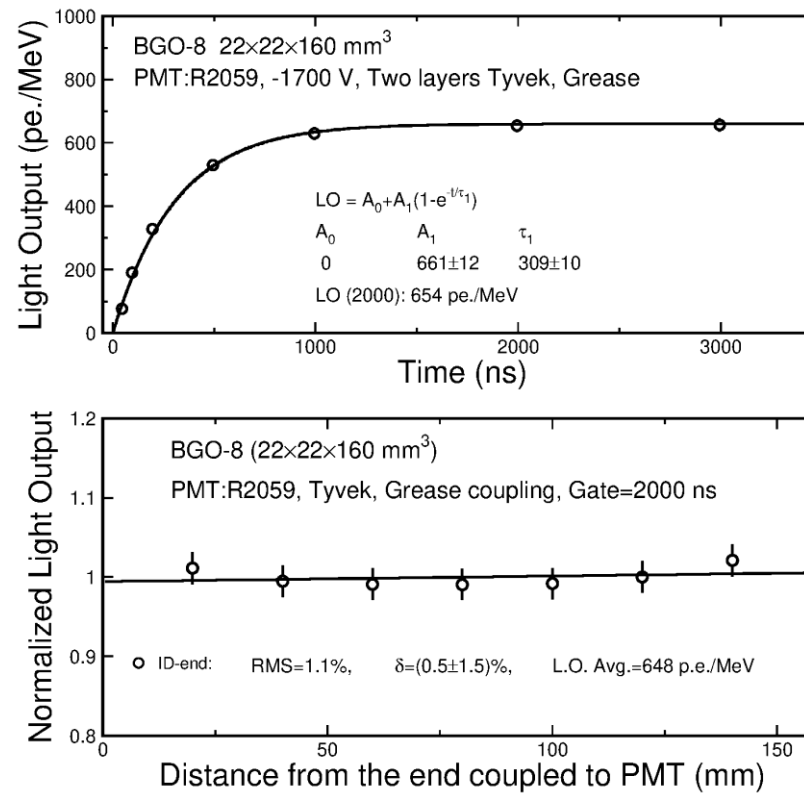
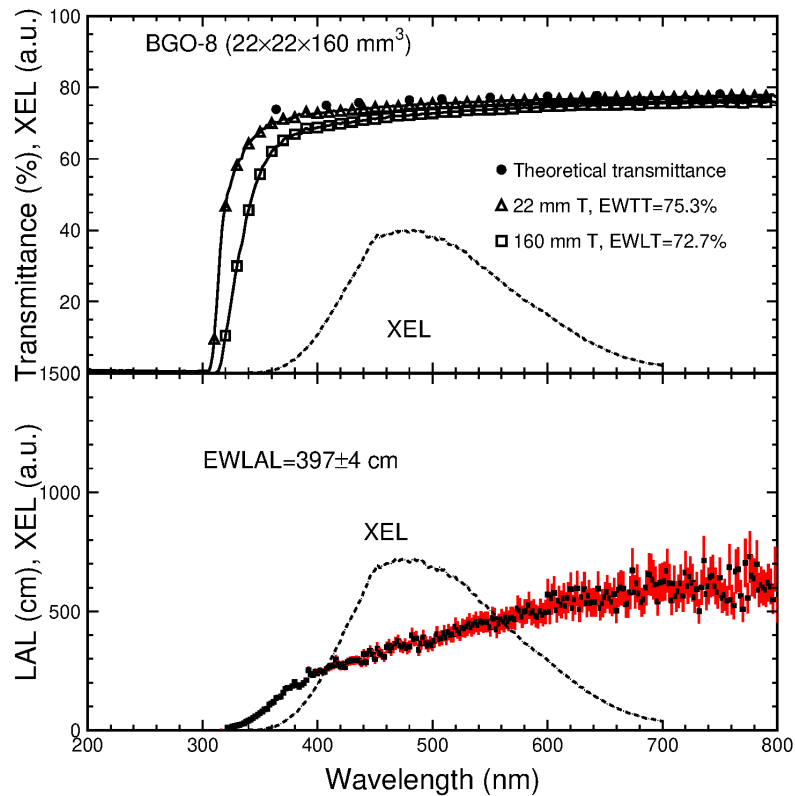
T, LAL, Decay, LO, LRU and ER (BGO-4)



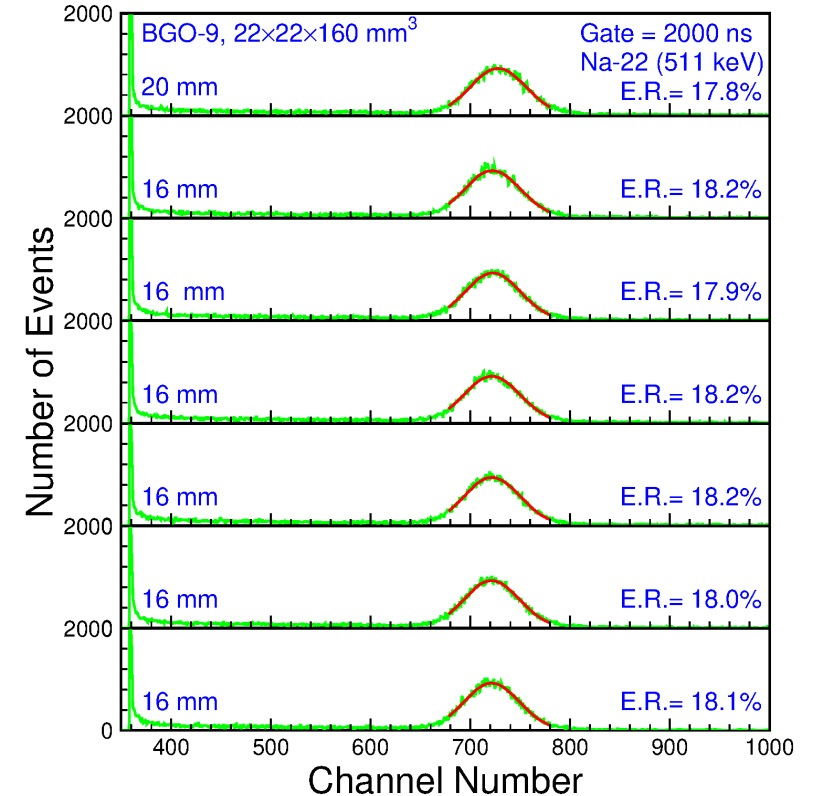
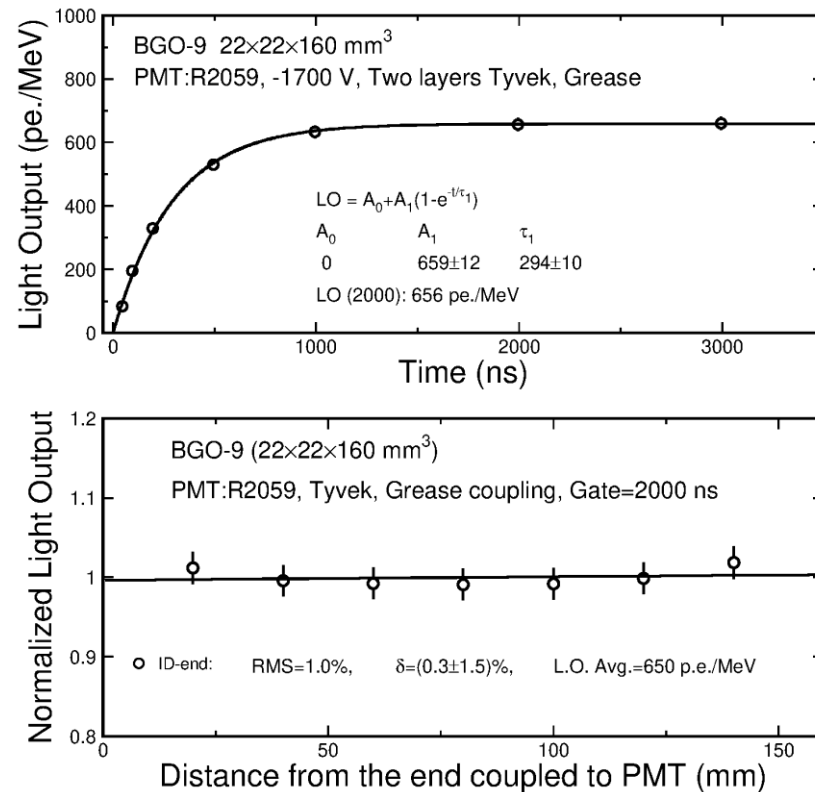
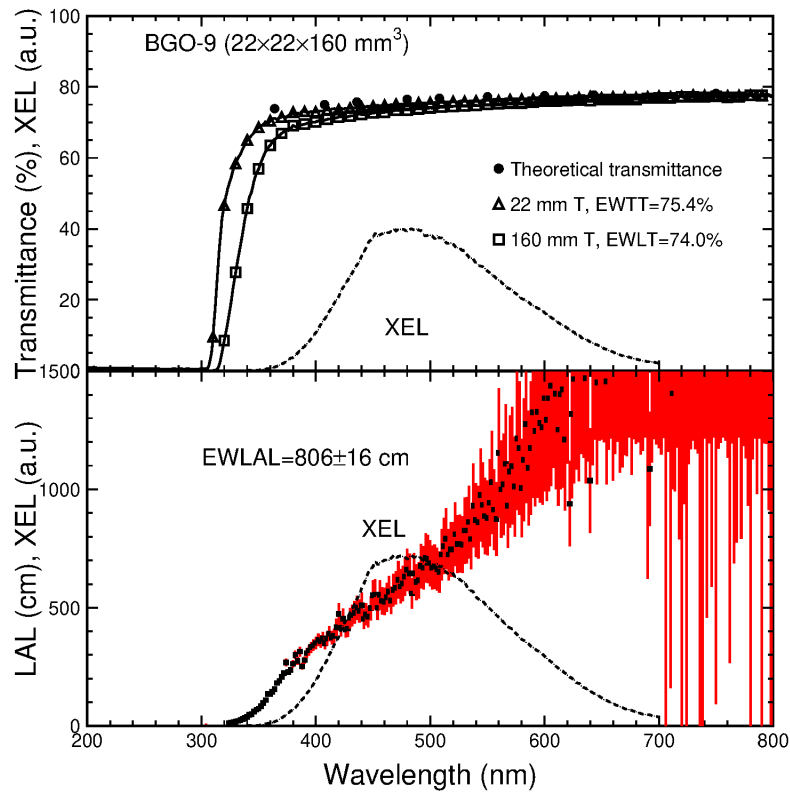
T, LAL, Decay, LO, LRU and ER (BGO-5)



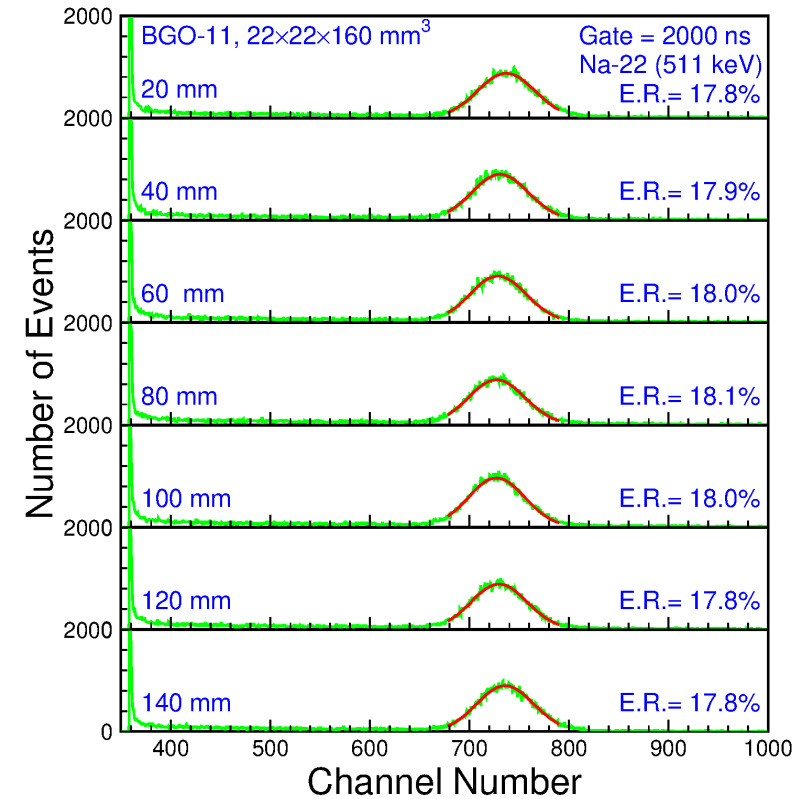
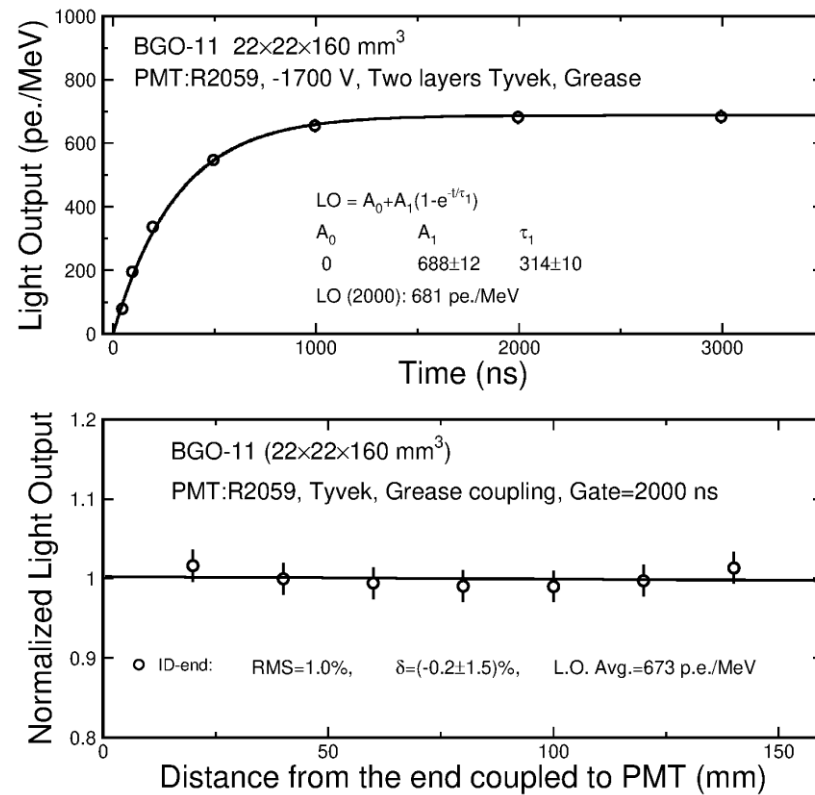
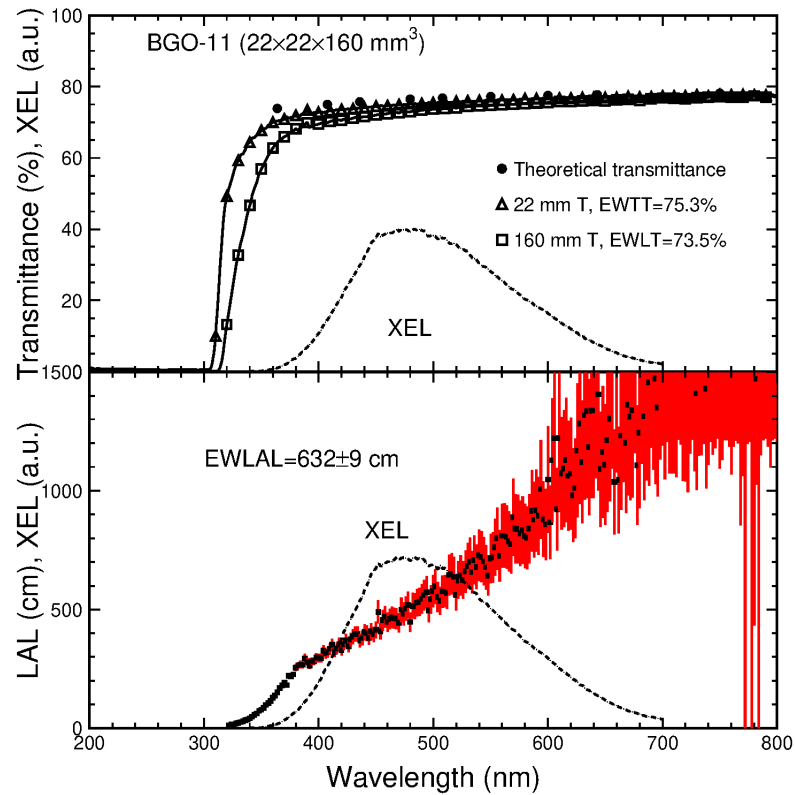
T, LAL, Decay, LO, LRU and ER (BGO-8)



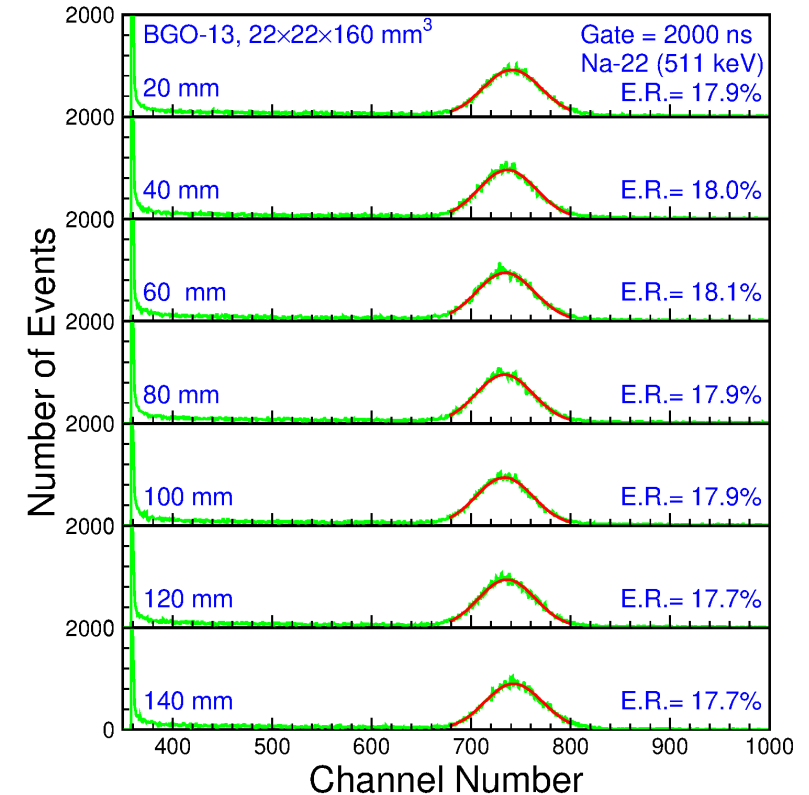
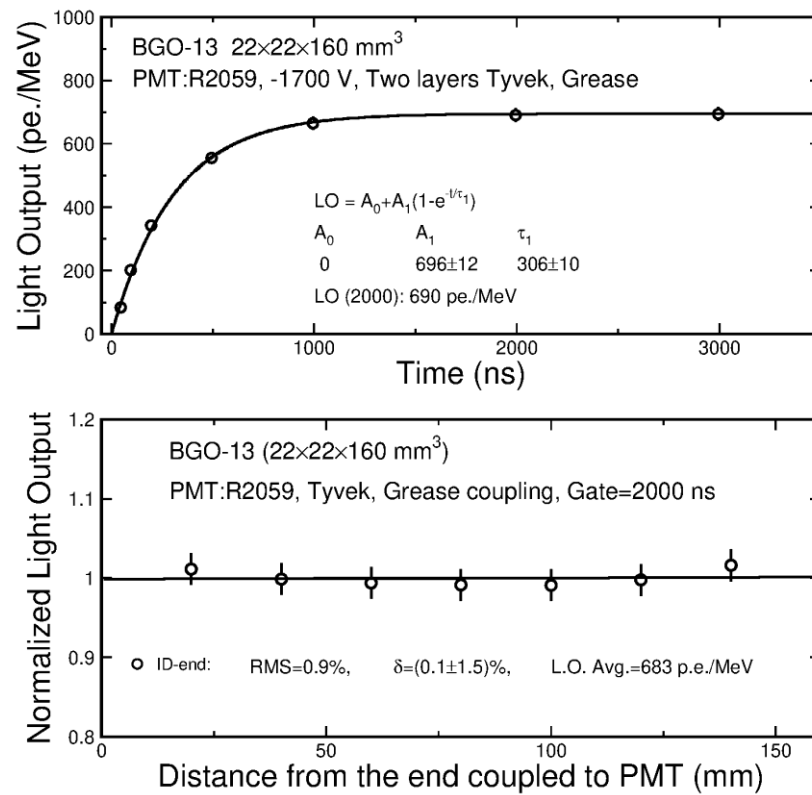
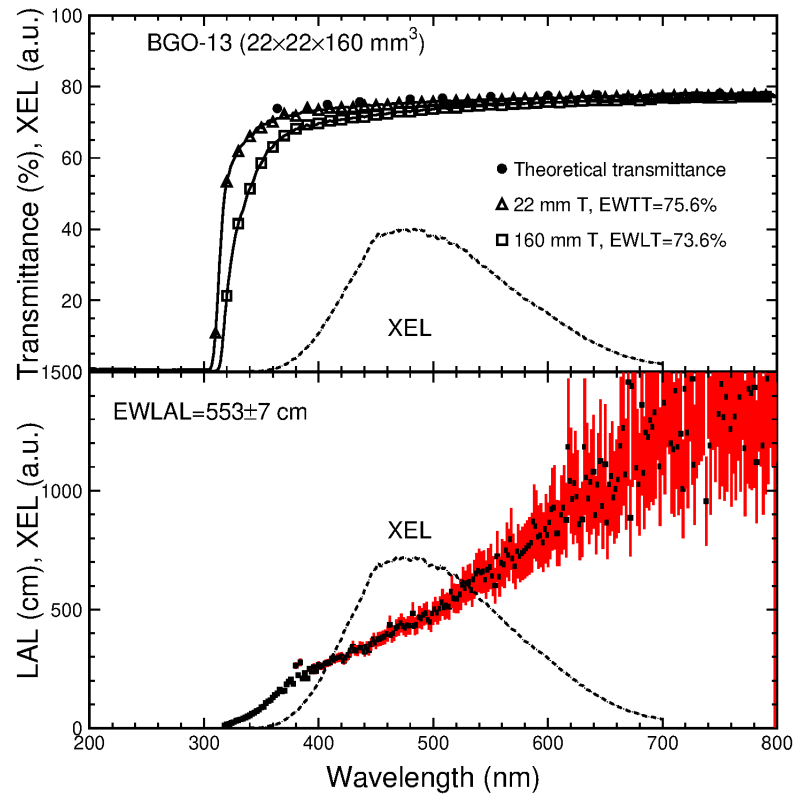
T, LAL, Decay, LO, LRU and ER (BGO-9)



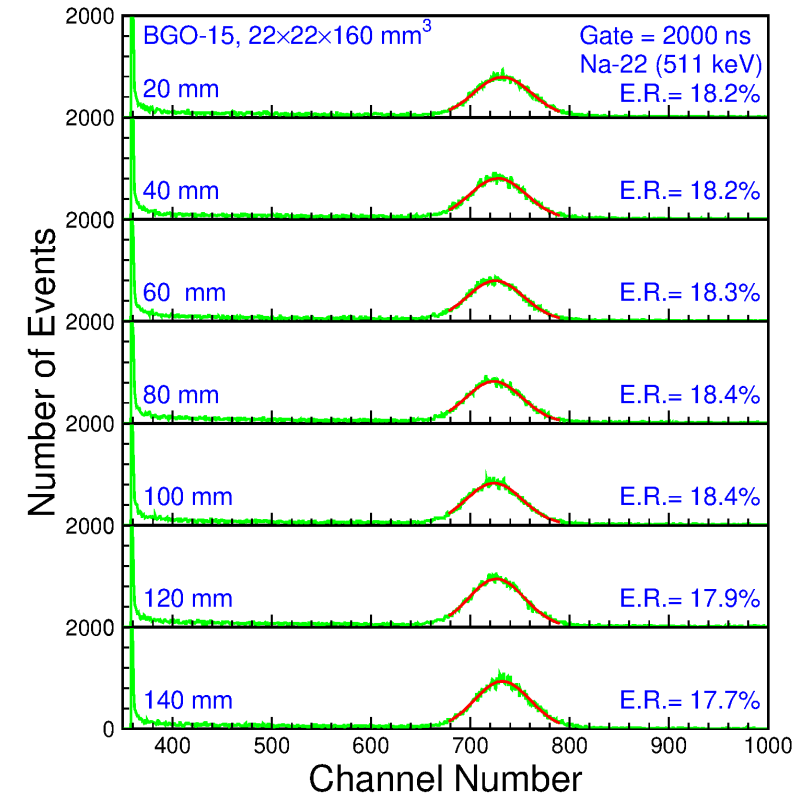
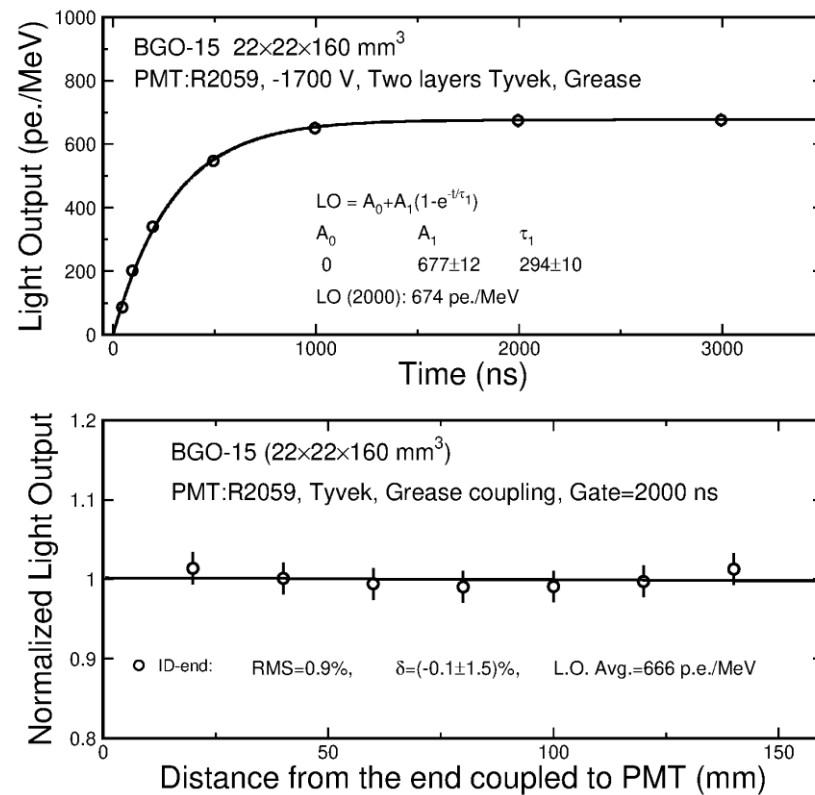
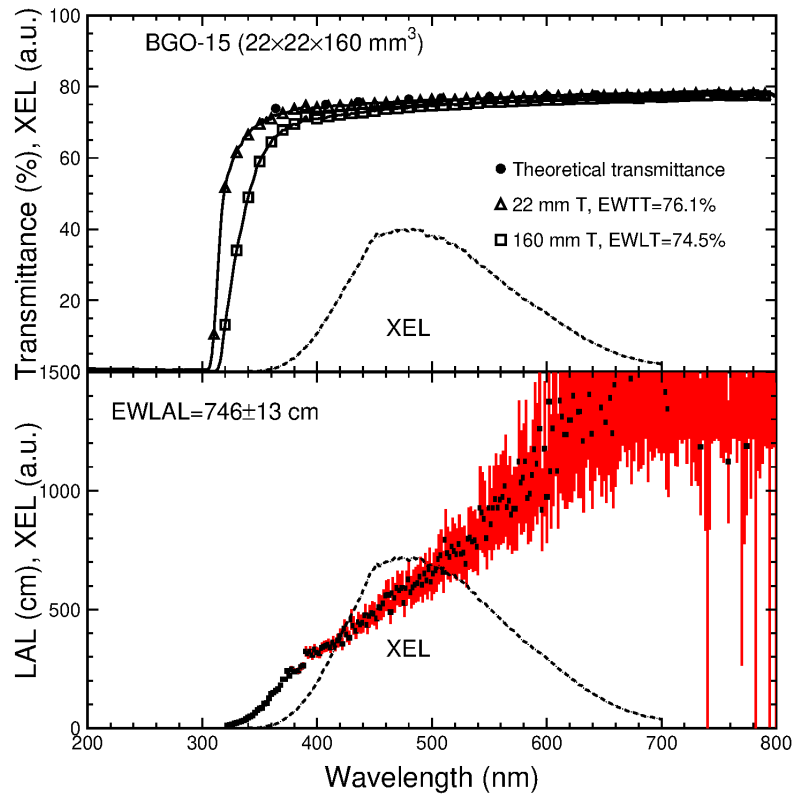
T, LAL, Decay, LO, LRU and ER (BGO-11)



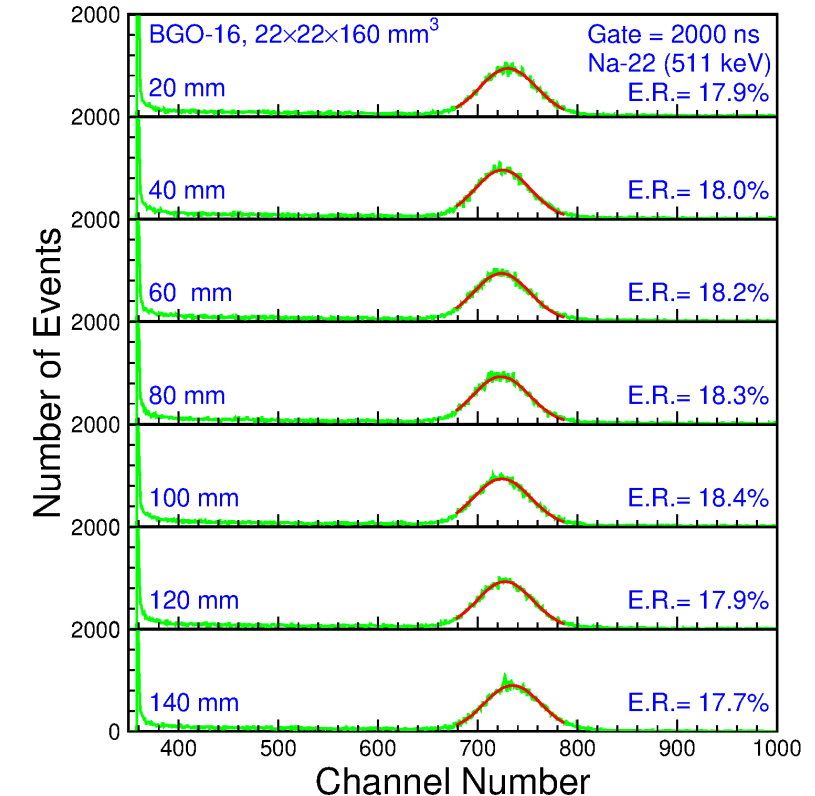
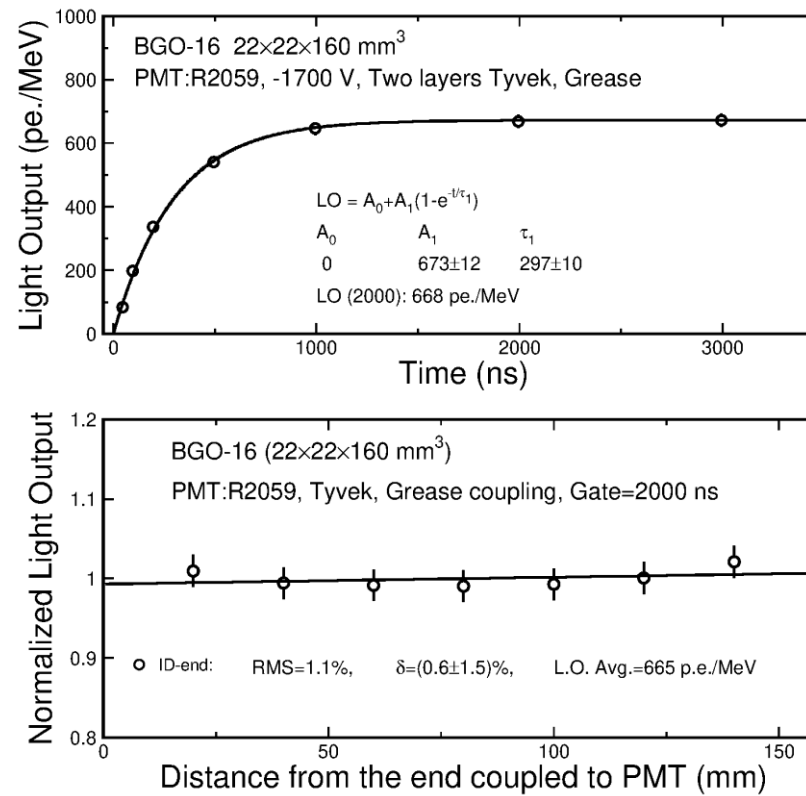
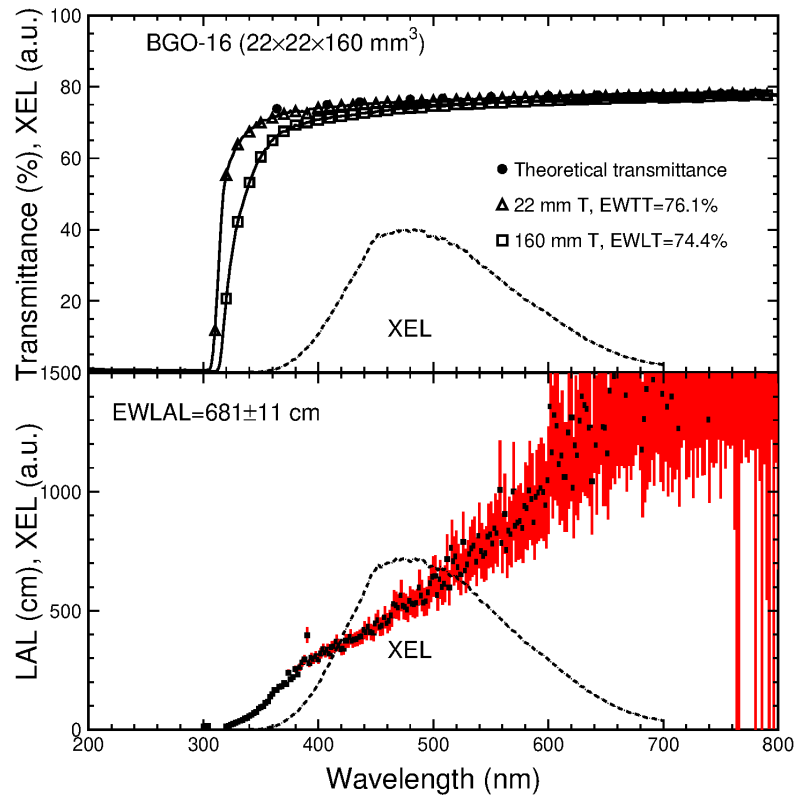
T, LAL, Decay, LO, LRU and ER (BGO-13)



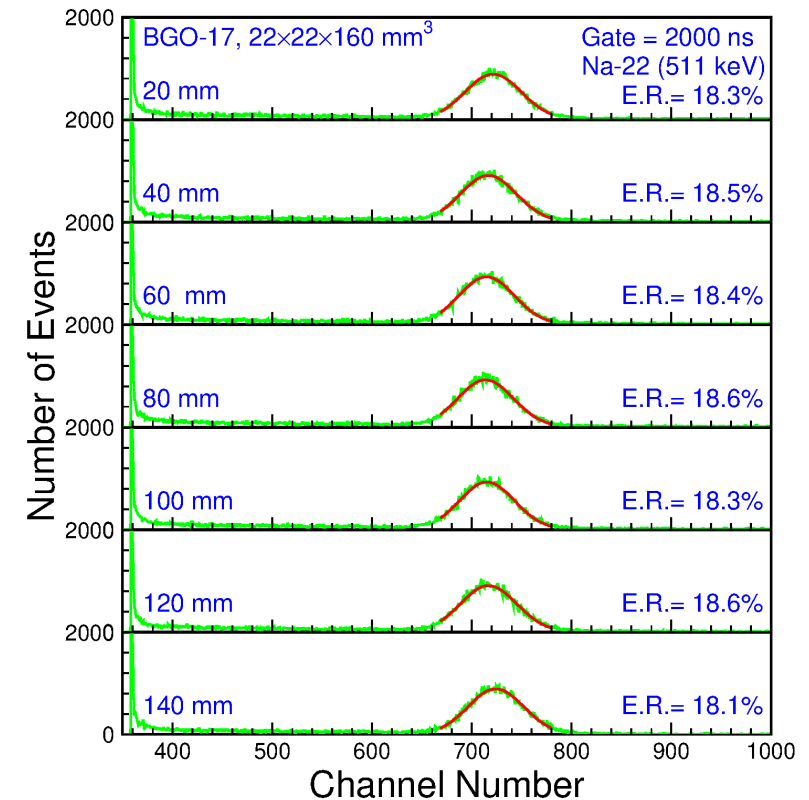
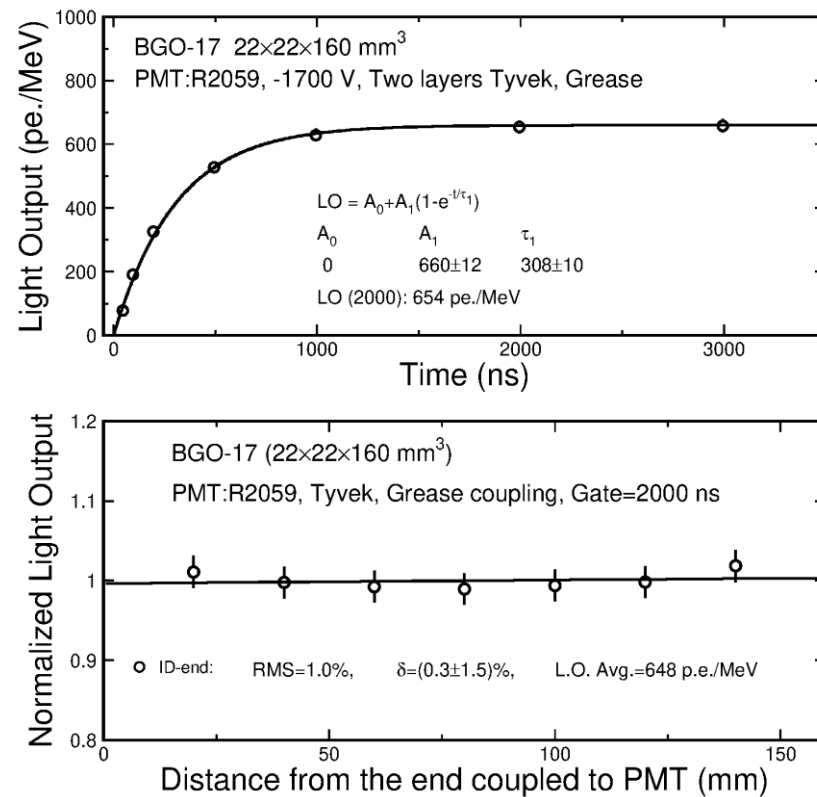
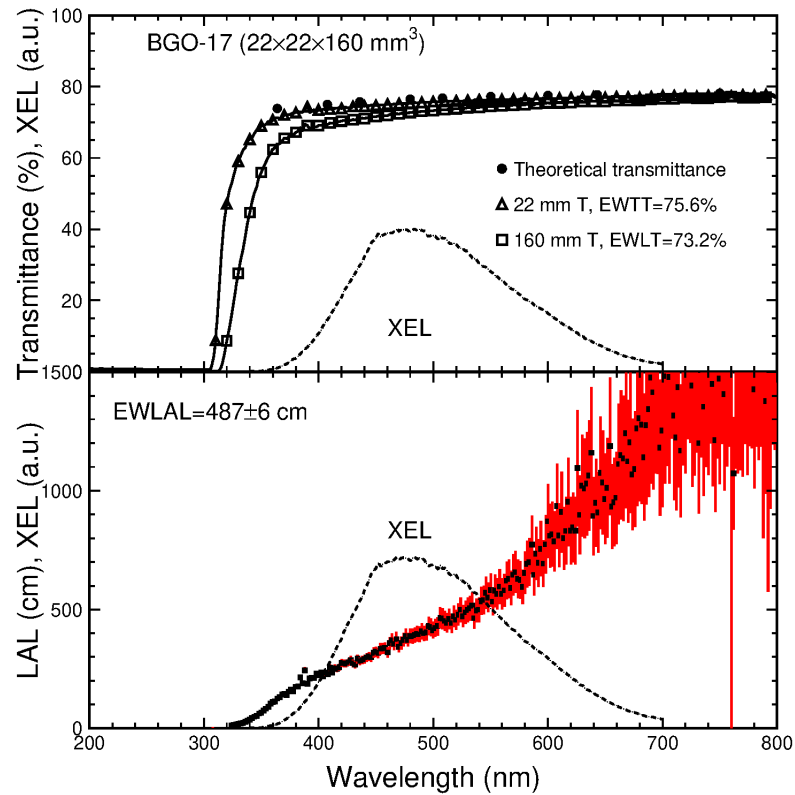
T, LAL, Decay, LO, LRU and ER (BGO-15)



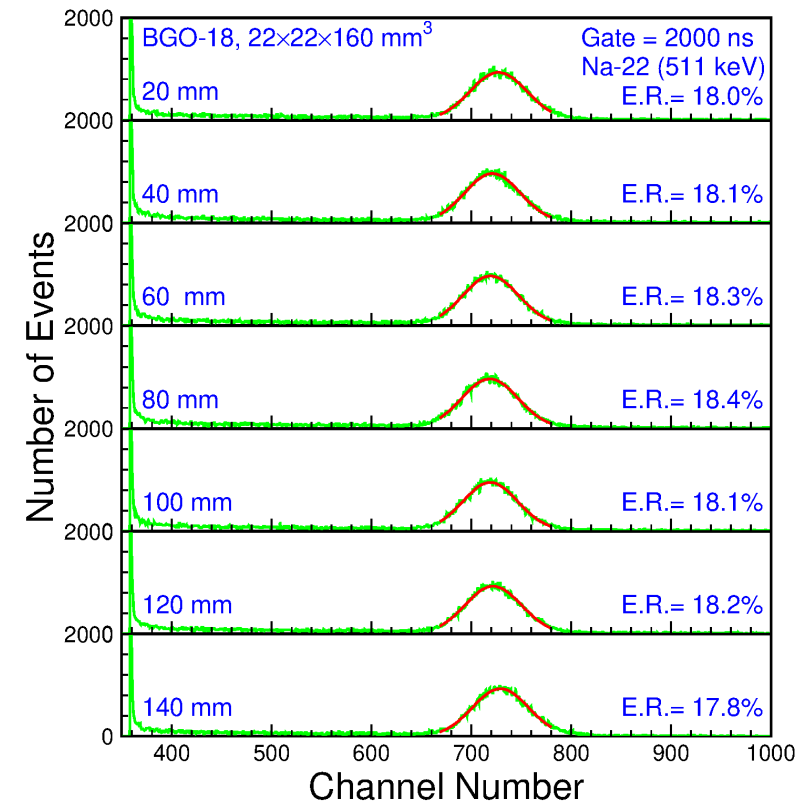
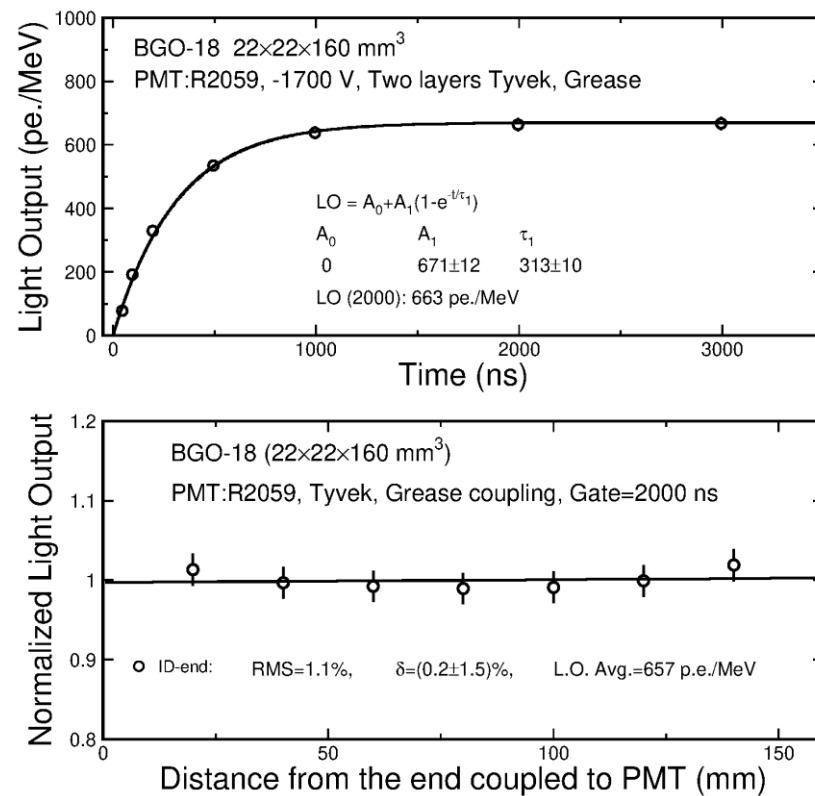
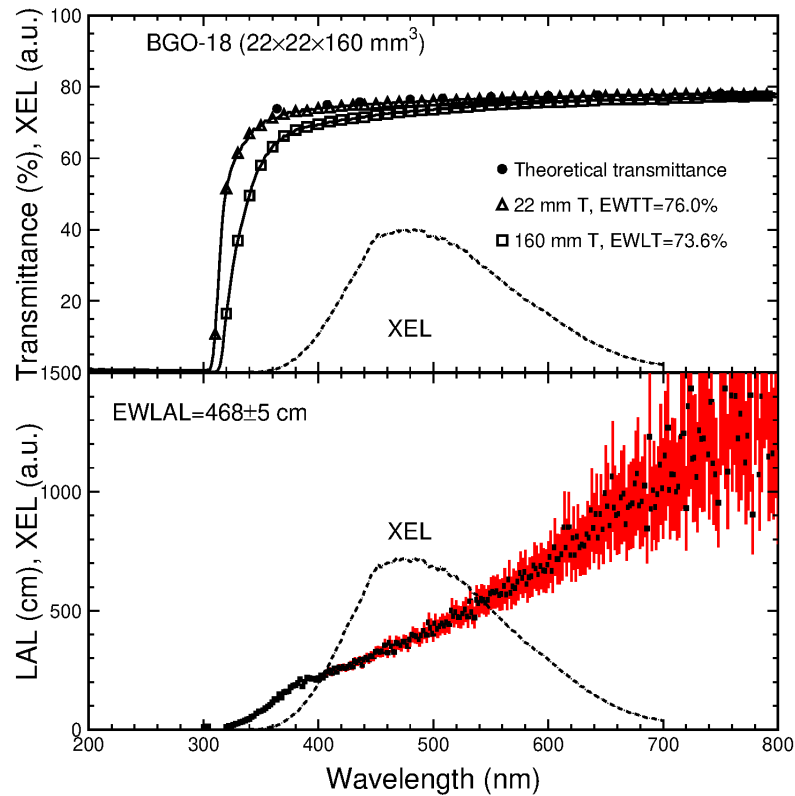
T, LAL, Decay, LO, LRU and ER (BGO-16)



T, LAL, Decay, LO, LRU and ER (BGO-17)



T, LAL, Decay, LO, LRU and ER (BGO-18)



T, LAL, Decay, LO, LRU and ER (BGO-19)

